Evidence for the numeric thresholds in the symbolic rule base

Domain-agnostic federated neuro-symbolic framework

Why each cut-point was chosen for Healthcare disease risk and Banking fraud

1. How the thresholds were chosen

Every symbolic rule in nsf/symbolic/rules.py fires on a fixed numeric condition (for example, systolic BP >= 140 or diastolic BP >= 90). None of these numbers were invented. They were fixed in two ways, and this document gives the source for each one so the choice can be checked independently.

- Healthcare: each threshold is a standard diagnostic or classification cut-point published by a recognised medical authority (WHO, the American Diabetes Association, NCEP ATP III, and respiratory-oxygen guidelines). These are the same numbers a clinician would use, so a rule firing has a defensible clinical meaning.
- Banking: there are no universal regulatory cut-points for fraud. The rules use the feature set and the directional logic that are well established in peer-reviewed fraud-detection research (amount, spend-vs-average ratio, geographic distance, transaction velocity, card-not-present, time-of-day, merchant risk). The specific numeric constants are engineering decision boundaries calibrated so that the symbolic layer agrees with the causal data generator that produced the dataset.
- In both domains the symbolic thresholds are aligned with the ground-truth generator in nsf/data/generators.py. That is why symbolic verification is meaningful on this data and not random: the rules test the same breakpoints that actually created the labels.

Honesty note for review: the healthcare numbers are externally grounded clinical standards. The banking numbers are literature-informed heuristics, not fixed values quoted from a paper. Section 4 states this explicitly so the distinction is clear.

2. Healthcare thresholds and their clinical sources

Task: classify a patient as low_risk, medium_risk, or high_risk. The table maps each rule's numeric condition to the published cut-point it reproduces, with the source key from Section 5.

Rule (code)	Condition in code	Clinical cut-point it reproduces	Src
hypertension	systolic >= 140 or diastolic >= 90 mmHg	Hypertension / Grade 1 HTN is office BP >= 140/90 mmHg (WHO 2021; ISH 2020).	H1, H2
hyperglycemia	glucose >= 126 mg/dL	Fasting plasma glucose >= 126 mg/dL (7.0 mmol/L) diagnoses diabetes (ADA 2026, Table 2.1).	H3
obesity	BMI >= 30	WHO defines obesity in adults as BMI >= 30 kg/m2 (overweight >= 25).	H4
hypoxia	oxygen_sat < 94 %	Normal target SpO2 is 94-98 %; below it, supplemental O2 is initiated and hypoxaemia considered.	H6
advanced_age	age >= 60	WHO treats 60+ as older adult; ATP III counts age as a CVD risk factor (M>=45/F>=55).	H5, H9
sedentary	exercise < 1.5 h/week	Well below the WHO minimum of 150 min (2.5 h)/week moderate activity.	H7
protective_exercise	exercise >= 5 h/week	At/above the upper WHO band of 150-300 min/week (5 h) linked to extra benefit.	H7

Rule (code)	Condition in code	Clinical cut-point it reproduces	Src
protective_sleep	sleep 7-9 h	National Sleep Foundation recommends 7-9 h/night for adults.	H8
smoker	smoker is true	Current smoking is an established, non-numeric CVD risk factor (ATP III Table 3).	H5
family_history	family_history is true	Family history of premature CHD is an established CVD risk factor (ATP III Table 3).	H5

Note on cholesterol: the dataset also records total cholesterol and the generator raises risk above 200 mg/dL, which matches the NCEP ATP III classification (< 200 desirable, 200-239 borderline high, >= 240 high). Cholesterol is used by the neural model rather than as its own firing rule, but its breakpoint is from the same ATP III source (H5).

2a. Why exactly 140 and 90 (the question raised)

- 140 mmHg systolic and 90 mmHg diastolic are the office blood-pressure values at which WHO (2021) recommends starting treatment for confirmed hypertension, and at which the 2020 International Society of Hypertension classifies Grade 1 hypertension (140-159 / 90-99 mmHg). The rule uses an OR because either an elevated systolic or an elevated diastolic reading meets the clinical definition.
- These are not arbitrary: the attached ISH 2020 guideline (Table 1) and the WHO 2021 recommendation both state the >= 140/90 boundary verbatim. The NCEP ATP III documents (attached) also list 'BP >= 140/90 mmHg' as a cardiovascular risk factor, which is why the same number appears across independent authorities.

3. Banking thresholds and their literature basis

Task: classify a transaction as legitimate or fraudulent. Unlike medicine, fraud detection has no universal numeric standard; thresholds are set per system. The table shows, for each rule, the feature and directional logic that is supported by the research literature, and marks the numeric value as a calibrated design choice.

Rule (code)	Condition	Feature / logic supported by literature	Src
high_amount	amount >= 900	Raw transaction amount; large amounts are a core fraud feature.	B1, B3
amount_spike	amount / 30-day avg >= 3.5	Spend-vs-customer-baseline ratio; behavioural aggregation of normal spend.	B1, B2
far_from_home	dist from home >= 80 km	Geolocation distance from the customer's usual location.	B2
geo_jump	dist from last txn >= 60 km	Distance between consecutive transactions / implausible travel speed.	B2
txn_burst	txns in 1 h >= 4	Transaction velocity: counts aggregated over a short recent window.	B1, B3
international	is_international is true	Country grouping is an established aggregation dimension.	B1
night_txn	hour in 00:00-05:59	Periodic time-of-day feature (von Mises periodic modelling).	B1
high_risk_merchant	merchant_risk >= 0.65	Merchant group / category risk is an aggregation dimension.	B1, B3
card_not_present	card_present is false	Transaction channel / type (CNP) is a standard fraud attribute.	B1
failed_pin	failed PIN attempts >= 2	Authentication-failure signal (operational/industry practice; weaker in academic literature).	-

4. Honest limitations (please read)

- The banking numeric constants (900, 3.5, 80 km, 60 km, 4, 0.65, 2 PIN attempts) are NOT quoted from any paper. They are decision boundaries chosen to match the causal fraud generator and to give a sensible score scale. The literature justifies WHICH features and WHICH direction, not the exact number.
- The healthcare cut-points are adult, general-population standards. They are not adjusted for sex, pregnancy, ethnicity, or measurement context (e.g., home vs office BP), which real clinical use would consider.
- The oxygen rule uses < 94 % for a soft risk signal and < 92 % for the hard override; 94-98 % is the normal target range and < 92 % is a common practical hypoxaemia threshold, so both are defensible but context matters.
- Guidelines are revised over time. For example, the 2017 ACC/AHA guideline lowers the US hypertension label to 130/80; this framework follows the WHO / ISH 140/90 convention and cites it explicitly.

5. References (attached in mail/sources unless marked link-only)

Healthcare

H1 WHO. Guideline for the pharmacological treatment of hypertension in adults. Geneva: WHO; 2021. BP threshold >= 140/90 mmHg. [link only] https://iris.who.int/ (CC BY-NC-SA 3.0 IGO).

H2 Unger T, et al. 2020 International Society of Hypertension Global Hypertension Practice Guidelines. Hypertension/J Hypertens. Grade 1 HTN 140-159/90-99 mmHg. [PDF attached]

H3 American Diabetes Association. 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2026. Diabetes Care 2026;49(Suppl 1):S27. FPG >= 126 mg/dL. [PDF attached]

H4 WHO. Obesity and overweight (fact sheet). Adults: overweight BMI >= 25, obesity BMI >= 30. [link only] https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight

H5 NCEP. Third Report of the Expert Panel (ATP III). NIH/NHLBI. Total cholesterol <200/200-239/>=240; risk factors include BP >= 140/90, smoking, family history, age. [PDF attached: At-A-Glance + Executive Summary]

H6 Thoracic Society of Australia & NZ, Acute Oxygen Guidelines 2016 (and BTS 2017 / NICE CKS). Target SpO2 94-98 %; < 92 % practical hypoxaemia threshold. [PDF attached]

H7 Bull FC, et al. WHO 2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med 2020;54:1451. 150-300 min/week moderate activity. [link only] https://bjsm.bmj.com/content/54/24/1451

H8 Hirshkowitz M, et al. National Sleep Foundation's updated sleep duration recommendations. Sleep Health 2015;1(1):40-43. Adults 7-9 h. [link only] https://doi.org/10.1016/j.sleh.2015.10.004

H9 WHO definition of older persons (60+ years), used together with the ATP III age risk factor (H5).

Banking

B1 Bahnsen AC, Aouada D, Stojanovic A, Ottersten B. Feature engineering strategies for credit card fraud detection. Expert Systems with Applications 2016;51:134-142. [PDF attached]

B2 Dal Pozzolo A, Caelen O, Le Borgne Y-A, Waterschoot S, Bontempi G. Learned lessons in credit card fraud detection from a practitioner perspective. Expert Systems with Applications 2014. [PDF attached]

B3 Whitrow C, Hand DJ, Juszczak P, Weston D, Adams NM. Transaction aggregation as a strategy for credit card fraud detection. Data Mining and Knowledge Discovery 2009;18:30-55. [PDF attached]